Math 110 — Week 9

Solutions

HW9 P1. Idea. The trace can be computed in any basis, so use one in which P is as simple as possible: P sends $\operatorname{null} P$ to 0 and fixes every vector of $\operatorname{range} P$.

1. $V = \operatorname{null} P \oplus \operatorname{range} P$. Every v is $(v - Pv) + Pv$, and $P(v - Pv) = Pv - P^2v = 0$, so $V = \operatorname{null} P + \operatorname{range} P$. If w is in both, then $w = Pu$ for some u and $Pw = 0$, so $w = Pu = P^2u = Pw = 0$; the sum is direct.
2. P fixes $\operatorname{range} P$: if $w = Pu$ then $Pw = P^2u = Pu = w$.
3. Take a basis $u_1, \dots, u_k$ of $\operatorname{null} P$ followed by a basis $w_1, \dots, w_r$ of $\operatorname{range} P$; together they are a basis of V by step 1. In this basis $Pu_i = 0$ and $Pw_j = w_j$, so the matrix of P is diagonal with k zeros and r ones.
4. So
$$\operatorname{tr} P = r = \dim \operatorname{range} P = \operatorname{rank} P.$$

HW9 P2. Idea. The minimal polynomial gives the size of the largest Jordan block for each eigenvalue (Week 8); the dimension 5 gives the total size. List the ways to fill 5, then read off the trace as the sum of the diagonal.

(a) 1. The roots of m_T are the eigenvalues, so they are 2 and -1.

2. The exponent of $x - 2$ is 2, so the largest block for 2 has size 2; the exponent of $x + 1$ is 1, so every block for -1 has size 1 (T6 of the handout).

3. Let a, b be the algebraic multiplicities of 2 and -1. Then $a \geq 2$ (there is a block of size 2), $b \geq 1$, and $a + b = 5$.

4. So (a, b) is $(2, 3)$, $(3, 2)$ or $(4, 1)$, and each occurs (part (c)).

(b) $\operatorname{tr} T = 2a + (-1)b$ (Prop. 79), so the possible traces are $4 - 3 = 1$, $6 - 2 = 4$ and $8 - 1 = 7$.

(c) $\operatorname{tr} T = 1$: $J_2(2) \oplus J_1(-1) \oplus J_1(-1) \oplus J_1(-1)$. $\operatorname{tr} T = 4$: $J_2(2) \oplus J_1(2) \oplus J_1(-1) \oplus J_1(-1)$. $\operatorname{tr} T = 7$: $J_2(2) \oplus J_2(2) \oplus J_1(-1)$, or $J_2(2) \oplus J_1(2) \oplus J_1(2) \oplus J_1(-1)$. In each, the largest block for 2 has size 2 and all blocks for -1 have size 1, so the minimal polynomial is $(x - 2)^2(x + 1)$.

HW9 P3. Idea. Trace is the sum of the eigenvalues, and a nilpotent operator has no eigenvalue other than 0.

(a) Let $N^k = 0$. If $Nv = \lambda v$ with $v \neq 0$, then $0 = N^k v = \lambda^k v$, so $\lambda^k = 0$ and $\lambda = 0$. So 0 is the only eigenvalue, with algebraic multiplicity $\dim V$, and $\operatorname{tr} N = 0$ by Prop. 79.

(b) $T = \operatorname{diag}(1, -1)$ has trace 0. It is not nilpotent: $T^2 = I$, so T^k is I or T, never 0.

(c) With $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$:
$$AB = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad BA = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad AB - BA = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \neq 0,$$
and $\operatorname{tr}(AB - BA) = 1 - 1 = 0$.

HW9 P4. Idea. All three are linear in each slot and symmetric, so only positivity needs checking. Look for a nonzero vector that makes $\langle x, x \rangle \leq 0$.

1. $\langle x, x \rangle_1 = x_1^2 + 2x_2^2$, which is positive unless $x_1 = x_2 = 0$. Linear in each slot and symmetric: an inner product.
2. $\langle (0, 1), (0, 1) \rangle_2 = -1 < 0$: not an inner product.
3. $\langle (1, -1), (1, -1) \rangle_3 = (1 - 1)^2 = 0$ although $(1, -1) \neq 0$: not an inner product.

HW9 P5. Idea. Linearity and symmetry come from the integral; positivity uses that a polynomial with infinitely many roots is 0. For (c), use symmetry of $[-1,1]$: the integral of an odd function is 0.

(a) The integral is linear, so $\langle \lambda p + r, q\rangle = \lambda\langle p, q\rangle + \langle r, q\rangle$; and $p(x)q(x) = q(x)p(x)$ gives symmetry. For positivity, $\langle p, p\rangle = \int_{-1}^{1} p^2$. The function p^2 is continuous and ≥ 0, so if the integral is 0 then $p^2 = 0$ on $[-1,1]$ by (T7). Then p has infinitely many roots, so p is the zero polynomial.

(b) $\|1\|^2 = \int_{-1}^{1} 1 = 2$, $\|x\|^2 = \int_{-1}^{1} x^2 = \frac{2}{3}$, $\|x^2\|^2 = \int_{-1}^{1} x^4 = \frac{2}{5}$. So the norms are $\sqrt{2}$, $\sqrt{2/3}$, $\sqrt{2/5}$.

(c) $\langle 1, x\rangle = \int_{-1}^{1} x = 0$ and $\langle x, x^2\rangle = \int_{-1}^{1} x^3 = 0$, because x and x^3 are odd. $\langle 1, x^2\rangle = \int_{-1}^{1} x^2 = \frac{2}{3} \neq 0$. So the orthogonal pairs are $\{1, x\}$ and $\{x, x^2\}$.

HW9 P6. Idea. $a + b + c$ is the dot product of (a, b, c) with $(1, 1, 1)$.

1. With $u = (a, b, c)$ and $v = (1, 1, 1)$ in $\mathbb{R}^3$, Cauchy–Schwarz gives
$$(a + b + c)^2 = \langle u, v\rangle^2 \leq \|u\|^2\|v\|^2 = 3(a^2 + b^2 + c^2).$$
2. Equality holds exactly when u, v are dependent. Since $v \neq 0$, that means $u = tv$ for some t, that is, $a = b = c$.

HW9 P7. Idea. Expand both squares; the cross terms appear with opposite signs.

1. Expand both by linearity in each slot:
$$\|u + v\|^2 = \langle u + v, u + v\rangle = \|u\|^2 + \langle u, v\rangle + \langle v, u\rangle + \|v\|^2,$$
$$\|u - v\|^2 = \langle u - v, u - v\rangle = \|u\|^2 - \langle u, v\rangle - \langle v, u\rangle + \|v\|^2.$$
2. Adding the two lines, the cross terms cancel and $\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2$. Nothing used the field, so this holds over $\mathbb{R}$ and over $\mathbb{C}$.

I1. Idea. $\operatorname{tr}(A^{\mathsf{T}}B)$ is the dot product of A and B viewed as lists of n^2 entries.

(a) Entry (j, j) of $A^{\mathsf{T}}B$ is row j of A^{T} (column j of A) times column j of B:
$$(A^{\mathsf{T}}B)_{jj} = \sum_i (A^{\mathsf{T}})_{ji} b_{ij} = \sum_i a_{ij} b_{ij},$$
so $\operatorname{tr}(A^{\mathsf{T}}B) = \sum_{i,j} a_{ij} b_{ij}$; with $B = A$ this is $\sum_{i,j} a_{ij}^2$. If $A^{\mathsf{T}}A = 0$, its trace $\sum a_{ij}^2$ is 0, so every $a_{ij} = 0$.

(b) By (a), $\langle A, B\rangle = \sum_{i,j} a_{ij} b_{ij}$ is the dot product on $\mathbb{R}^{n^2}$, with the entries listed in some order. So it is linear in A, symmetric, and $\langle A, A\rangle = \sum a_{ij}^2 > 0$ for $A \neq 0$.

(c) This is Cauchy–Schwarz for this inner product, squared: $\langle A, B\rangle^2 \leq \langle A, A\rangle\langle B, B\rangle$.

I2. Idea. A real matrix's non-real eigenvalues come in conjugate pairs with equal multiplicity (handout, "Real matrices"), and there are only three eigenvalues in all.

1. A is real, so $1 - i$ is also an eigenvalue, with the same multiplicity as $1 + i$ (handout, "Real matrices"). Together with 2 and $1 + i$ that is three eigenvalues, and the multiplicities add up to 3, so each has multiplicity 1.
2. $\operatorname{tr} A = 2 + (1 + i) + (1 - i) = 4$.
3. Yes. Three distinct eigenvalues give an eigenbasis, so $A = SDS^{-1}$ with $D = \operatorname{diag}(2, 1 + i, 1 - i)$, and $A^{-1} = SD^{-1}S^{-1}$. So
$$\operatorname{tr} A^{-1} = \operatorname{tr} D^{-1} = \frac{1}{2} + \frac{1}{1+i} + \frac{1}{1-i} = \frac{1}{2} + \frac{(1-i) + (1+i)}{2} = \frac{3}{2}.$$

I3. Idea. A commutator always has trace 0; that rules out (b) at once, and (a) has trace 0, so try matrix units.

1. (a) Take $A = E_{12}$, $B = E_{21}$. Then $AB = E_{11}$ and $BA = E_{22}$, so $AB - BA = \operatorname{diag}(1, -1)$ (as computed in HW9 P3(c)).
2. (b) For all A, B, $\operatorname{tr}(AB - BA) = 0$ (Prop. 80). The matrix $\operatorname{diag}(1, 0)$ has trace $1 \neq 0$, so no A, B give it.

I4. Idea. An average is an inner product with a constant vector; $\int xf$ is the L^2 product of f with x.

(a) With $u = (x_1, \dots, x_n)$ and $v = (1, \dots, 1)$, Cauchy–Schwarz gives $(\sum x_i)^2 \le n \sum x_i^2$. Divide by n^2:

$$\left(\frac{x_1 + \cdots + x_n}{n}\right)^2 \le \frac{x_1^2 + \cdots + x_n^2}{n}.$$

(b) In the L^2 product on $[0,1]$, $\int_0^1 xf(x)\,dx = \langle f, x\rangle$ and $\|x\|^2 = \int_0^1 x^2\,dx = \frac{1}{3}$. Cauchy–Schwarz gives $\langle f, x\rangle^2 \le \frac{1}{3}\|f\|^2$, which is the inequality. Equality holds exactly when f and x are dependent: $f(x) = cx$ for a constant c (including $c = 0$).

I5. Idea. Polarization recovers the inner product from the norm: expand $\frac{1}{4}(\|u+v\|^2 - \|u-v\|^2)$.

1. With $N(x,y) = x^2 + xy + y^2$, the terms of $N(u+v) - N(u-v)$ that do not cancel are the cross terms:

$$\frac{N(u+v) - N(u-v)}{4} = u_1v_1 + \tfrac{1}{2}u_1v_2 + \tfrac{1}{2}u_2v_1 + u_2v_2.$$

2. This is linear in u and symmetric in u, v. Positivity:

$$\langle u, u\rangle = u_1^2 + u_1u_2 + u_2^2 = (u_1 + \tfrac{1}{2}u_2)^2 + \tfrac{3}{4}u_2^2,$$

which is 0 only if $u_2 = 0$ and then $u_1 = 0$. So it is an inner product, and $\langle u, u\rangle = N(u)$ shows it gives the norm.

I6. Idea. The angle comes from $\cos\theta = \langle u, v\rangle/(\|u\|\,\|v\|)$; only the length of $(1, \dots, 1)$ changes with n.

1. $\langle (1,\dots,1), e_1\rangle = 1$, $\|(1,\dots,1)\| = \sqrt{n}$ and $\|e_1\| = 1$, so by Def. 45

$$\cos\theta = \frac{1}{\sqrt{n}}, \qquad \theta = \arccos\frac{1}{\sqrt{n}}.$$

2. As $n \to \infty$, $1/\sqrt{n} \to 0$, so $\theta \to \pi/2$: in high dimension the diagonal is almost perpendicular to every coordinate axis.

I7. Idea. Weights depend only on differences of scores.

(a) $\langle (0,1), k_j\rangle = 0, 1, 0$, so the scores are 0, $1/\sqrt{2}$, 0, and the weights are

$$\frac{1}{2 + e^{1/\sqrt{2}}},\ \frac{e^{1/\sqrt{2}}}{2 + e^{1/\sqrt{2}}},\ \frac{1}{2 + e^{1/\sqrt{2}}} \approx 0.248,\ 0.504,\ 0.248.$$

The query is perpendicular to k_1 and k_3, so those two tie.

(b) For $q = (t, 0)$ the scores are $t\sqrt{2}$, 0, $-t/\sqrt{2}$. Divide the numerator and denominator of a_1 by $e^{t\sqrt{2}}$:

$$a_1 = \frac{1}{1 + e^{-t\sqrt{2}} + e^{-3t/\sqrt{2}}} \longrightarrow 1 \quad (t \to \infty),$$

and $a_2, a_3 \to 0$ because the weights add up to 1. (At $t = 3$ the weights are already 0.984, 0.014, 0.002.)

(c) $\langle q, k_j + w\rangle = \langle q, k_j\rangle + \langle q, w\rangle$: every score increases by the same number $\langle q, w\rangle/\sqrt{d_k}$, and softmax does not change when the same number is added to all scores. Doubling the keys doubles the scores to $2\sqrt{2}$, 0, $-\sqrt{2}$, whose differences are twice as large, and the weights become about 0.932, 0.055, 0.013.